

Nichols algebras and root systems – Exercises

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ABSTRACT. These are the exercises of Heckenberger’s course **Nichols Algebras and Root Systems**, given at the Vrije Universiteit Brussel, Faculty of Sciences, Department of Mathematics and Data Sciences, as part of the Francqui VUB-Leerstool 2025–2026.

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Introduction

This document contains the exercises for the 20-hour course Nichols Algebras and Root Systems, given in February and March 2026 at the Vrije Universiteit Brussel. The course is based on the book [1].

Several people attending this course helped write or improve exercises from the lectures, including Silvia Properzi, Lukas Simons, and Leandro Vendramin.

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§ 1. Braided structures

1.1. EXERCISE. Prove that the category of vector spaces is a monoidal category.

1.2. EXERCISE. Let $\mathcal{C}$ be a monoidal category and $\mathcal{C}^{\text{op}}$ its dual category of $\mathcal{C}$. Prove that $\mathcal{C}^{\text{op}}$ is a monoidal category.

1.3. EXERCISE. Let $\mathcal{C}$ be a monoidal category.

- (a) Prove that the unit of an algebra A in $\mathcal{C}$ is unique.
- (b) Prove that the counit of a coalgebra C in $\mathcal{C}$ is unique.

1.4. EXERCISE. Let $\mathcal{C}$ be a monoidal category, (C, Δ, ϵ) a coalgebra in $\mathcal{C}$, and (A, μ, η) an algebra in $\mathcal{C}$. Prove that the convolution product of $\text{Hom}_{\mathcal{C}}(C, A)$ is associative with unit $\eta\epsilon$.

1.5. EXERCISE. Let $(\mathcal{C}, \otimes, I, c)$ be a strict braided monoidal category. Prove that

$$c_{I,X} = c_{X,I} = \text{id}$$

for all $X \in \mathcal{C}$.

1.6. EXERCISE. Let $(\mathcal{C}, \otimes, I, c)$ be a braided strict monoidal category. Prove that $\mathcal{C}^{\text{op}}$ is a braided strict monoidal category.

1.7. EXERCISE. Let $(\mathcal{C}, \otimes, I, c)$ be a braided strict monoidal category. Prove that $\bar{\mathcal{C}}$ is a braided strict monoidal category.

1.8. EXERCISE. Let $\mathcal{C}$ be a strict monoidal category, A, B, C and D algebras in $\mathcal{C}$, and $\varphi: A \rightarrow C$ and $\psi: B \rightarrow D$ be algebra morphisms in $\mathcal{C}$. Let (V, λ_A) be a left A -module in $\mathcal{C}$ and (W, λ_B) be a left B -module in $\mathcal{C}$. Prove the following statements:

- 1) $A \otimes B$ is an algebra in $\mathcal{C}$ with multiplication

$$\mu_{A \otimes B} = (\mu_A \otimes \mu_B)(\text{id} \otimes c_{B,A} \otimes \text{id})$$

and unit $\eta_{A \otimes B} = \eta_A \otimes \eta_B$.

- 2) $\varphi \otimes \psi: A \otimes B \rightarrow C \otimes D$ is an algebra morphisms in $\mathcal{C}$.
- 3) $V \otimes W$ is an $A \otimes B$ -module with $(\lambda_A \otimes \lambda_B)(\text{id} \otimes c_{B,V} \otimes \text{id})$.
- 4) The algebra structures of $(A \otimes B) \otimes C$ and $A \otimes (B \otimes C)$ coincide.

1.9. EXERCISE. Formulate and prove the dual version of Exercise 1.8.

1.10. EXERCISE. Let $\mathcal{C}$ be a strict braided monoidal category and H a bialgebra in $\mathcal{C}$.

- (a) Prove that the category ${}_H\mathcal{C}$ of left H -modules in $\mathcal{C}$ is a strict monoidal category.
- (b) Prove that the category ${}^H\mathcal{C}$ of left H -comodules in $\mathcal{C}$ is a strict monoidal category.

1.11. EXERCISE. Let K be a field and G be a group. Prove that a Yetter–Drinfeld module over KG is a G -graded KG -module $V = \bigoplus_{g \in G} V_g$ such that $g \cdot V_h \subseteq V_{ghg^{-1}}$ for all $g, h \in G$.

1.12. EXERCISE. Let $\mathcal{C}$ be a braided monoidal category. Define for every $(Y, \lambda) \in {}_H\mathcal{C}$ and $(X, \delta) \in {}^H\mathcal{C}$

$$c_{X,Y}^{\mathcal{YD}(\mathcal{C})} = (\lambda \otimes \text{id})c_{X,Y}(\delta \otimes \text{id}_Y).$$

Let $(V, \lambda) \in {}_H\mathcal{C}$ and $(V, \delta) \in {}^H\mathcal{C}$. Prove that the following statements are equivalent:

- 1) $(V, \lambda, \delta) \in {}_H^H\mathcal{YD}(\mathcal{C})$.
- 2) For all $X \in {}_H\mathcal{C}$, the map $c_{V,X}^{\mathcal{YD}(\mathcal{C})}$ is a morphism in ${}_H\mathcal{C}$.

- 3) $c_{V,H}^{\mathcal{YD}(\mathcal{C})}$ is a morphism in ${}^H\mathcal{C}$.
 4) For all $X \in {}^H\mathcal{C}$, the map $c_{X,V}^{\mathcal{YD}(\mathcal{C})}$ is a morphism in ${}^H\mathcal{C}$.

1.13. EXERCISE. Let H be a Hopf algebra in $\mathcal{C}$ with invertible antipode. Prove that ${}^H_H\mathcal{YD}(\mathcal{C})$ is a strict braided monoidal category with monoidal structure as for modules and comodules and the braiding

$$c_{X,Y}^{\mathcal{YD}(\mathcal{C})} = (\lambda_Y \otimes \text{id})c_{X,Y}(\delta_X \otimes \text{id}_Y)$$

for $X, Y \in {}^H_H\mathcal{YD}(\mathcal{C})$.

§ 2. Nichols algebras

2.1. EXERCISE. Let $\mathcal{C}$ be a strict braided monoidal category. Prove the universal property of the tensor algebra in $\text{gr}_{\mathbb{N}_0}\mathcal{C}$.

2.2. EXERCISE. Let $\mathcal{C}$ be a strict braided monoidal category and $V \in \mathcal{C}$. Prove that the tensor algebra $T(V)$ is a Hopf algebra with bijective antipode in $\text{gr}_{\mathbb{N}_0}\mathcal{C}$.

2.3. EXERCISE. Let $\mathcal{C}$ be a strict braided monoidal category, $V \in \mathcal{C}$ and A an algebra in $\mathcal{C}$.

- (a) Prove that for every morphism $f : V \rightarrow P_{\eta,\eta}(A)$ there is a unique bialgebra morphism $\varphi : T(V) \rightarrow A$ such that $\varphi|_V = f$.
 (b) Show that, if A is in $\text{gr}_{\mathbb{N}_0}\mathcal{C}$ and $f(V) \subseteq P_{\eta,\eta}(A(1))$, then φ is a morphism in $\text{gr}_{\mathbb{N}_0}\mathcal{C}$.

2.4. EXERCISE. Let V be a complex braided vector space with basis $\{x_1, x_2\}$ and braiding

$$c(x_i \otimes x_j) = q_{ij}x_j \otimes x_i, \quad i, j \in \{1, 2\}.$$

Compute the Nichols algebra of (V, c) in the following cases:

- 1) $q_{11} = q_{22} = -1$ and $q_{12}q_{21} = 1$.
 2) $q_{11} = q_{22} = -1$ and $q_{12}q_{21} = -1$.
 3) $q_{11} = q_{22} = -1$ and $q_{12}q_{21} = \omega$, where ω is a primitive cube root of unity. This Nichols algebra is usually denoted by $\mathcal{U}_q(\mathfrak{sl}_3)^+$. Prove that in this algebra the elements $y^{n_1}(\text{ad}_c x)(y)^{n_2}x^{n_3}$ span $\mathcal{U}_q(\mathfrak{sl}_3)^+$.

§ 3. Weyl groupoids

3.1. EXERCISE. Let $\mathcal{G} = \mathcal{G}(I, \mathcal{X}, r, A)$ be a semi-Cartan graph. Let $i_1, \dots, i_k \in I$, $X, Y, Z \in \mathcal{X}$, $w \in \text{Hom}(X, Y)$ and $w' \in \text{Hom}(Y, Z)$. Prove the following statements:

- 1) $|l(w) - l(w')| \leq l(w'w) \leq l(w') + l(w)$ and $l(w^{-1}) = l(w)$.
 2) $l(w'w) \equiv l(w') + l(w) \pmod{2}$.
 3) $l(s_i w) \in \{l(w) + 1, l(w) - 1\}$ and $l(ws_i) \in \{l(w) + 1, l(w) - 1\}$.
 4) $k - l(\text{id}_X s_{i_1} \cdots s_{i_k})$ is non-negative and even.

3.2. EXERCISE. Let $\mathcal{G} = \mathcal{G}(I, \mathcal{X}, r, A)$ be a Cartan graph. Let $X \in \mathcal{X}$ and $i \in I$. Assume that both $\Delta^{X\text{re}}$ and $\Delta^{r_i(X)\text{re}}$ are contained in $\mathbb{N}_0^I \cup -\mathbb{N}_0^I$. Prove the following statements:

- 1) $s_i^X(\pm\alpha_i) = \mp\alpha_i$ and $s_i^X(\Delta_+^{X\text{re}} \setminus \{\alpha_i\}) = \Delta_+^{r_i(X)\text{re}} \setminus \{\alpha_i\}$.
 2) $m_{ij}^X = m_{ij}^{r_i(X)}$.

3.3. EXERCISE. Let $\mathcal{G} = \mathcal{G}(I, \mathcal{X}, r, A)$ be a Cartan graph, $X, Y \in \mathcal{X}$ and $w \in \text{Hom}(Y, X)$. Prove the following statements:

- 1) $N(w) = N(w^{-1})$.
- 2) Assume that both $\Delta^{Y^{\text{re}}}$ and $\Delta^{r_i(Y)^{\text{re}}}$ are contained in $\mathbb{N}_o^I \cup -\mathbb{N}_0^I$. If $w(\alpha_i) \in \mathbb{N}_0^I$, then $N(ws_i) = N(w) + 1$. If $w(\alpha_i) \in -\mathbb{N}_0^I$, then $N(ws_i) = N(w) - 1$.

References

- [1] I. Heckenberger and H.-J. Schneider. *Hopf algebras and root systems*, volume 247 of *Math. Surv. Monogr.* Providence, RI: American Mathematical Society (AMS), 2020.